

Natalia Rabinovich

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EDUCATION

Rutgers University Honors College

New Brunswick, NJ

Bachelor of Science in Aerospace Engineering, Dean's List

May 2028

- **Relevant Coursework:** Fluid Mechanics, Thermodynamics, Mechatronics, CAD, Calc I – V
- **Technical Training:** UNP Satellite Fabrication Course

RELEVANT EXPERIENCE

Program Manager

Sep 2024 – Present

SPICESat CubeSat, Rutgers

New Brunswick, NJ

- Lead an 8-subteam, 40-member CubeSat program, coordinating mechanical, thermal, electrical, and software teams through spacecraft design, hardware integration, testing, and major industry-level technical reviews.
- Own Phase A execution within UNP's spacecraft development cycle, tracking engineering requirements, action items, schedules, technical documentation, and hardware risks through spacecraft design reviews.
- Lead weekly multidisciplinary design reviews to identify subsystem interface conflicts, assign action owners, and resolve technical risks before hardware testing, submission deadlines, and major program reviews.

Lead Systems Engineer

Fall 2026

NASA L'SPACE Mission Concept Academy

Online

- Selected as Lead Systems Engineer for NASA L'SPACE Mission Concept Academy, leading requirements development, subsystem interface definition, trade studies, and system-level technical coordination.

Undergraduate Research Assistant

Fall 2026

Space Propulsion with Advanced Chemistry and Energetics (SPACE) Lab, Rutgers

New Brunswick, NJ

- Supporting research in chemical and electric propulsion, advanced propellants, and VLEO propulsion systems.

Thermal Engineering Project Lead

Summer 2026

SPICESat CubeSat, Rutgers

New Brunswick, NJ

- Led a 4-member team developing the thermal-control strategy for a CubeSat fluid payload, translating 9 mission requirements into thermal limits, orbital analysis cases, and hardware design constraints for CDR readiness.
- Simplified spacecraft CAD in SolidWorks for thermal modeling by removing noncritical hardware and small geometric features while preserving key geometry, interfaces, contact surfaces, and component dimensions.
- Ran 10+ hot/cold case orbital simulations in Thermal Desktop and use MATLAB to process temperature histories, predicting payload temperatures from -10C to 40C, and improving fluid temperature distribution gradient by 20%.

Mechanical Lead

Sep 2022 – May 2024

Questionable Engineering Team, FIRST Robotics Competition

Jersey City, NJ

- Led mechanical design in Onshape/AutoCAD, achieving reliable arm motion and object manipulation in matches.

PROJECTS

Autonomous Docking Robot

Spring 2026

Mechatronics Project

- Designed, built, and tested an autonomous docking robot using SolidWorks, 3D-printed hardware, an HC-SR04 sensor, DC motors, LEDs, and a servo-actuated clasp, using sensor to control speed and trigger docking sequence.

Mechanical Claw Machine

Spring 2026

SolidWorks CAD Project

- Designed a complete claw-machine assembly in SolidWorks, including the casing, gantry, claw, access door, shafts, guides, gears, and rack-and-pinion components needed for horizontal travel, vertical extension, and claw actuation.

SKILLS & AWARDS

Mechanical Design: SolidWorks, Onshape, AutoCAD, Fusion, FEA, CAD, 3D printing, mechanical prototyping

Hardware: Hand tools, soldering, wiring, staking, potting, conformal coating, cleanroom practices, hardware integration

Engineering: Trade studies, mechanical integration, technical documentation, PDR/CDR preparation, test planning

Programming: MATLAB, Python, C++, HTML, CSS, JavaScript

Software: Thermal Desktop, Microsoft Excel, PowerPoint, Word, Git, Linux

Languages: Fluent in English, Spanish, and Russian